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Local Agglomeration and Household Mortgage Debt

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Abstract. Using detailed household data, we find that households working in locally agglomerated economies have high mortgage loans and are more likely to have mortgage loans. Channel testing shows that local agglomeration increases housing and mortgage demand as well as mortgage supply. The increase in mortgage loans is driven by purchases of larger houses and smaller down payments. Mechanism analyses document that local agglomeration affects mortgage loans by increasing career upward potential and providing downside protection. Specifically, we find that the impact of local agglomeration is stronger for skilled employees with enhanced prospects. In addition, local agglomeration weakens the negative relation between unemployment and mortgage loans, which supports the downside-protection role of local agglomeration. These results hold under instrumental variables analysis, difference-in-difference analysis, and a set of robustness checks. Overall, our findings highlight the importance of local labor market composition in household mortgage debt.

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Keywords: local agglomeration • mortgage debt • labor market composition

1. Introduction

Mortgage debt is the largest component of household debt, which plays a key role in driving economic fluctuations (Mian et al. 2017).¹ Although previous studies have examined the dynamics of mortgage debt in the United States, especially after the 2008–2009 financial crisis, the analyses are mainly from the perspective of credit expansion (see Mian and Sufi 2009, Agarwal et al. 2014, Adelino et al. 2016) and regulatory interventions (Defusco et al. 2019, Chakraborty et al. 2020). Scarce research considers geographical factors as potential sources of the heterogeneity in household mortgage debt. In this paper, we examine whether local agglomeration affects household mortgage loans. Our investigation is built on and motivated by recent work documenting the important role of geographical factors in individuals' economic decisions (e.g., Chetty et al. 2014, Finkelstein et al. 2016, Addoum et al. 2024).

Local agglomeration, also known as an agglomerated economy, is defined as a group of firms in the same industry being clustered in a geographic location. The earliest concept of local agglomeration dates back to a theoretical analysis of the emergence of agglomeration economies by Marshall (1890).² Subsequent theoretical

and empirical studies suggest that agglomerated economies increase corporate productivity, stimulate innovative partnerships, and present opportunities for entrepreneurial activity (Porter 1998, Almazan et al. 2010, Dougal et al. 2015, Engelberg et al. 2018). However, agglomerated economies also carry costs and risks. First, they involve fierce competition, which may lower firm survival rates (Valta 2012, Dougal et al. 2015). Second, they can exacerbate negative externalities by contaminating neighboring firms' performance and subsequently accelerating the bankruptcy process (Benmelech et al. 2019).

In contrast to the ample documentation of the impact of local agglomeration on corporate activities, its impact on households is largely ignored in previous literature. Agglomerated economies affect local labor market conditions, which is an important factor for both borrowers and lenders in considering mortgage loans. Specifically, the characteristics of local agglomeration, such as knowledge spillovers and fierce competition, significantly affect the career risks of employees. Literature and anecdotal evidence indicate that such risk impacts mortgage loan decisions. For example, borrowers who expect a safe future are more likely to take out higher

levels of mortgage debt (Cocco 2013). In addition, mortgage lenders are more willing to lend to employees who have lower default rates (Hsu et al. 2018). Moreover, anecdotes imply that lenders worry about applicants' career risk. For example, the U.S. mortgage broker, Gustan Cho Associates, states that "lenders want to see the continuity of employment. Lenders also want to see the likelihood of future employment so they do not have trouble paying their mortgage payments timely."³ Further, Rocket Mortgage, the largest retail mortgage lender in the United States, suggests that mortgage lenders care about applicants' employment stability and career potential: "lenders like to see a 2-year history in your current job position. However, it's possible to be given the green light without that 2-year history if you're transferring into a new role."⁴ In this regard, the labor market condition aligns well with the framework used by lenders to evaluate applicants.

Theoretically, the link between local agglomeration and household mortgage loans is ambiguous, depending on the perspective taken. The career prospects view suggests a positive relationship between local agglomeration and mortgage loans. Career prospects affect mortgage loans through increasing upward potential and providing downside protection. From the upside view, local agglomeration increases the potential of employees in labor markets by improving their skills. In particular, the proximity to knowledgeable individuals facilitates the acquisition of skills and the exchange and diffusion of knowledge (Glaeser 1999), leading to the so-called knowledge spillover view in agglomerated economies.⁵ For example, spying, imitation, and rapid interfirm movement of highly skilled labor among neighboring firms in Silicon Valley led to the rapid dissemination of ideas related to the development of computer chips. The opportunities for acquiring better skills allow individuals to have stable (or even better) careers in the future. From the downside-protection view, local agglomeration provides protection for laborers in agglomerated economies. Workers benefit from local agglomeration, as they have a variety of job opportunities, making it easier to find alternative jobs in case of job loss. In this case, the frictions between job search and recruiting are reduced by local agglomeration (Krugman 1991). Duranton and Puga (2004) argue that as the workforce and the number of firms grow, the average worker can find a better match for their skills. Altogether, the career prospects view suggests that local agglomeration increases employees' upward potential and provides downside protection for laborers. These risk-based effects give households higher expectations for the future, making them more assertive in housing markets and mortgage markets (Gerardi et al. 2010). Meanwhile, local agglomeration creates a favorable environment for lenders with regard to assessing borrowers' career potential and employment stability.

Alternatively, the layoff risk view suggests that workers in clustered industries are less likely to have mortgage loans. First, firms in agglomerated economies are highly competitive, increasing firms' bankruptcy rate and laborers' layoff risk. In particular, Dougal et al. (2015) argue that firms in locally agglomerated economies compete with one another to grab the market. The fierce competition increases firms' business risks and reduces their survival rate (Valta 2012). Thus, households working in the area could be exposed to higher layoff risks. Second, the geographical proximity of firms in agglomerated economies exacerbates the negative externalities. More precisely, the economies of agglomeration can be detrimental during downturns, propagating and amplifying the negative effects of financial distress and bankruptcies among firms in the same locality. As an example, Benmelech et al. (2019) find that bankrupt firms impose negative externalities on non-bankrupt neighboring firms, causing contagion to spread from financially distressed companies in agglomerated economies. Third, local agglomeration has been criticized for overspecialization (Granovetter 1985), which further increases the layoff risk. Overspecialization makes firms prone to rather rigid strategies with respect to technological and market potential, with the eventual evolution into a closed system that is vulnerable to downturns (Glaeser et al. 1992). Under this condition, firms' marginal return is stagnant and even decreases, and laborers face high layoff risk.⁶ Thus, higher layoff risk makes households less likely to apply or get approval for mortgages. The layoff risk view therefore predicts a negative link between local agglomeration and household mortgage loans.

We use data provided by the Current Population Survey (CPS) to capture local agglomeration. Following previous literature (Holmes 2005, Holmes and Stevens 2014, Addoum et al. 2024), we define local agglomeration as the labor supply share of an industry in a local labor market scaled by its nationwide labor supply share. Our measure successfully captures well-known agglomerated economies such as the industry of hotels and motels in Las Vegas, Nevada. We use the Survey of Income and Program Participation (SIPP) to capture household mortgage loans, and our sample period spans from 1984 to 2003.

We conduct a household-level analysis and find empirical results supporting the career prospects view. Specifically, we document that households that work in agglomerated economies have a larger value of mortgage loans, as well as a higher probability of holding mortgage loans. Regarding the economic significance, an increase of one standard deviation in local agglomeration is associated with a 2% increase in mortgage loans. Given that the average value of household mortgage debt is \$41,213, an increase of one standard deviation in local agglomeration leads to a dollar increase in

household mortgage loans of \$824. The effect of local agglomeration is economically important and comparable to other studies on household mortgage debt (Barrot et al. 2022).

In robustness checks, we extend our sample period to alleviate the concern that the data underlying our analyses are too obsolete to allow generalization of our findings to current mortgage markets. We employ two different methods to expand the sample period. The first method relies on the Home Mortgage Disclosure Act (HMDA) data set, which provides application-level data since 1990 and allows us to use a sample period extending from 1990 to 2022. The second method involves reconstructing our local agglomeration measure at the state-industry-year level, which enables us to extend our SIPP sample to 2011 (SIPP does not provide the state information in recent years). Utilizing these two methods, we find that the positive association between local agglomeration and mortgage loans continues to hold. In addition, our results are robust to using alternative fixed effects and excluding extremely large labor markets or leading industries.

A potential argument is that endogeneity because of reverse causality or omitted variables could bias our results. We address this concern in two ways. First, we use local IPO activity as an exogenous shock to local agglomeration and then validate our findings using a difference-in-differences (DiD) research design. As public firms have more production activities and hire more workers than private firms (Jiang et al. 2024), local IPO activity increases local agglomeration. Our DiD analysis shows that local IPO activity (proxied by the emergence of the first intensive IPO wave) increases household mortgage loans. Second, we instrument local agglomeration using the United States's granting of permanent normal trade relations (PNTRs) to China in 2001. This event removes the uncertainty of China's most favored nation (MFN) status, increases the import competition from Chinese firms, and leads to significant employment losses for U.S. firms (Autor and Dorn 2013, Pierce and Schott 2016). Specifically, we instrument local agglomeration, using an indicator for tradable sectors in which production could have been outsourced to Chinese competitors in the post-PNTRs period after 2001. We find that the positive link between local agglomeration and mortgage loans continues to hold.

Given the relationship between local agglomeration and mortgage loans, we next examine the underlying channels. First, we find that local agglomeration significantly increases housing demand, proxied by the mortgage loan amount in applications and in approvals (Agarwal et al. 2022) and hedonic-adjusted house value (Han 2010). In addition, local agglomeration increases mortgage demand, proxied by the number of mortgage loan applications. Second, we find that local agglomeration increases mortgage

supply, as measured by approval rate and the number of granted loans.⁷ Finally, we find that households in agglomerated economies are more likely to have (more) mortgage loans because they purchase larger houses and simultaneously because they have smaller down payments.

Next, we examine the mechanisms behind the career prospects view. First, we find that the impact of local agglomeration on mortgage loans is more pronounced for skilled employees. Given that Acemoglu and Autor (2011) and Addoum et al. (2024) suggest that skilled employees have better prospects for promotions and careers being enhanced, this result supports our claim of the upside potential role of local agglomeration. In addition, we find that risky individuals (i.e., unemployed individuals) are more likely to obtain mortgages in more agglomerated economies than in less agglomerated economies. This outcome is because of the thick labor market of local agglomeration, which optimizes the process of matching employees to positions and reduces unemployment spells. These cross-sectional analyses suggest that local agglomeration increases the career prospects of employees and serves as a protective factor for individuals who are perceived to be risky. Overall, these results lend support to the arguments that the career prospects channel drives the increase in mortgage debt through risk-based effects.

Next, we rule out alternative explanations for the link between local agglomeration and mortgage debt. First, the possibility exists that households select locations based on their positive views on agglomerated economies. To mitigate such selection concern, we examine the link between local agglomeration and migration. Using total inflow and net inflow of employees to measure migration, we find no significant association between local agglomeration and migration. Second, greater housing supply and rising house prices in agglomerated economies may potentially drive our results (Mian and Sufi 2011).⁸ To test this alternative explanation, we control for the housing supply elasticity and house price growth, and our main result still holds.

Finally, we conduct additional analyses. First, we find that households in local agglomerations earn more and accumulate more wealth. This finding supports the wealth effect of local agglomeration. It also lends support to the theoretical argument that laborers in agglomerations earn more because of enhanced skills. Second, we find that local agglomeration significantly increases payment years, whereas it has no significant impact on monthly mortgage payments. Third, we consider the impact of firm size, finding that local agglomeration plays a role for households regardless of whether they are working for either small or big firms. Finally, we examine whether the effect of agglomerated economies varies across industries. We find that local agglomeration significantly increases mortgage loans for

households working in the manufacturing industry, whereas it has no significant impact on other industries.

This paper contributes to the mortgage debt literature by highlighting the importance of location-based factors, namely, local agglomeration. Until now, the focus of mortgage debt literature has mainly been on person-based factors or macroeconomic factors. These factors include borrowers' unemployment (Hsu et al. 2018), individual trust (Jiang and Lim 2018), credit supply (Mian et al. 2020), house price (Mian and Sufi 2011), and interest rate structure (Kojien et al. 2009). To the best of our knowledge, our paper is the first to discuss the impact of the local agglomerative pattern on mortgage debt. More importantly, this geographic characteristic varies across households even within a local area, distinguishing our findings from the well-documented local housing market effect (Mian and Sufi 2009).⁹ Moreover, we show that the underlying channel behind the increased mortgage loans goes beyond the wealth effect, and we highlight the risk-based role of local agglomeration. That is, local agglomeration increases upside potential and provides downside protection.

This paper also contributes to the literature of local agglomeration by documenting its impact on household finance. Existing local agglomeration literature predominantly focuses on corporate behavior, with very little discussion of the impact on households. For example, Kedia and Rajgopal (2009) examine whether the location of a firm's headquarters explains variation in broad-based option grants. Relatedly, Almazan et al. (2010) suggest that firms located within agglomerated economies have more acquisition opportunities, and Dougal et al. (2015) find that a firm's investment is highly sensitive to the investments of firms headquartered nearby. Additionally, Davis et al. (2014) report a positive link between agglomerated economies and total factor productivity, and Engelberg et al. (2018) find that firms in industry clusters have more efficient stock market prices than firms outside clusters. Against the backdrop of prior research that seldom analyzes whether location-based factors affect household behaviors, we fill a research gap by documenting a positive relation between agglomeration economies and household mortgage debt. Close to our work, Addoum et al. (2024) examine the impact of industries' geographic location on household portfolio choice. Our paper furthers their work by showing that the location-based factor (i.e., local agglomeration) affects mortgage debt.

2. Data and Variables

2.1. Sample Construction

To investigate the effect of local agglomeration on household mortgage debt, we start with data provided by SIPP (Chetty et al. 2017, Célerier and Matray 2019).¹⁰ SIPP is a household-based survey designed as a continuous series of national panels. Each panel generally

features a large sample of households that are interviewed multiple times over a four-year period. In particular, SIPP collects detailed information about household debts and demographics from 20,000 to 30,000 households over several (eight to 12) waves within each panel. Each wave includes a "core" survey that collects household sociodemographic data and several topical modules that gather specific information on a wide variety of subjects, such as work disability history, education and training history, and fertility history. We use the core survey data to obtain information about household demographics and the assets and liabilities topical module to collect data on household debts. We start our sample from 1984, the year SIPP began. Notably, SIPP has not provided metropolitan statistical area (MSA) information since its 2004 panel (Corradin and Popov 2015). As local agglomeration is constructed at the MSA level (Addoum et al. 2024), our sample period ends with 2003.

Given the criticism of SIPP's imputation methodology, we follow Gruber and Yelowitz (1999) and Célerier and Matray (2019) by dropping all observations with imputed wealth information. In addition, we limit our sample to households whose heads are between 24 and 65 years old, which reflects the typical pattern of individuals entering the labor market at the age of 24 and retiring at 65 in the United States (Krebs et al. 2015). Next, we construct our sample in a pooled cross-sectional structure because local agglomeration is more likely to affect the original amount of mortgage loans.¹¹ Specifically, we keep observations when households' first mortgage loan is originated. If these data are not available, we keep the observation in first year when the household was surveyed.¹² Finally, we merge the SIPP data with the data of local agglomeration. Our final sample includes 41,366 household-level observations.

2.2. Measuring Mortgage Loans

As previously mentioned, the data on mortgage debt are obtained from SIPP (Chetty et al. 2017, Célerier and Matray 2019). In particular, we focus on a household's first mortgage loan because it is more likely to be driven by housing demand, whereas second and third mortgages are probably driven by investment need. We use two measures to gauge the first mortgage loan. First, we take a logarithmic transformation of the value of mortgage loans at origination ($\text{Log}(1+mdebt)$) because Célerier and Matray (2019) argue that wealth and asset variables have highly positive skewness. Second, we use a dummy variable (Dum_mdebt) to indicate whether a household has the first mortgage loan.

2.3. Measuring Local Agglomeration

To calculate local agglomeration, we download data from the CPS. The CPS is a monthly U.S. household survey conducted jointly by the U.S. Census Bureau and the Bureau of Labor Statistics on over 65,000 households

from 1976 to the present. These surveys gather information on education, labor force status, demographics, and other aspects of the U.S. population. Following Addoum et al. (2024), when calculating local agglomeration, we restrict the sample to workers aged between 16 and 64 and laborers who work more than 35 hours per week and 40 weeks per year. In addition, we use sampling weights when aggregating individual labor supply to the industry level. We use MSAs to calibrate the local labor market.

Specifically, we identify the level of local agglomeration by referring to the measure of the location quotient statistic commonly used in prior literature (Holmes 2005, Holmes and Stevens 2014, Addoum et al. 2024). The detailed definition of local agglomeration of industry j in local labor market m is as follows:

$$Local\ aggl_{j,m,t} = \frac{S_{jmt}}{S_{jMt}},$$

where S_{jmt} is industry j 's labor supply in local labor market (MSA) m , scaled by the total labor supply of all the industries in labor market m in year t . Put differently, S_{jmt} is the labor share of industry j in local labor market m in year t . S_{jMt} is industry j 's labor supply across the whole country scaled by the total labor supply of all industries in the United States in year t . Meanwhile, we define the labor supply as the product of the number of weeks worked in the previous year and the usual hours worked per week in the previous year (Addoum et al. 2024). As such, if the value of the local agglomeration variable is larger than one, it indicates that industry j is highly concentrated in the local labor market.

To show the validity of our measure of local agglomeration, we list the 15 most locally agglomerated industry-location pairs in Table A.1 of Appendix A. Several notable industry-MSA pairs appear on the list. For example, Las Vegas, Nevada, well-known for tourism and entertainment, ranks first for its hotel and motel industry, with an agglomeration level of 14.350. Meanwhile, Detroit, Michigan, nicknamed the Motor City, ranks second in the list, having an agglomeration value of 11.763. Fresno, California, makes the list because of its predominant large-scale agricultural production, with a value of 10.104. In addition, some other MSAs rank highly because of their extremely localized industries, including Greensboro-Winston Salem, North Carolina; Austin, Texas; and Washington, DC/Maryland/Virginia.

2.4. Control Variables

Following previous literature on household debt, we control for four groups of control variables (Hsu et al. 2018, Célerier and Matray 2019). The first group incorporates household sociodemographic information, including household size and number of children.

Specifically, *Size* is the number of individuals in a household, and *Num_kid* is the number of children in a household. The second group of controls is related to household heads' basic information, including marital status, gender, education level, age, and employment status. More precisely, *Married* is a dummy variable equal to one if the household head is married, and zero otherwise; *Female* is also an indicator equal to one if the household head is female, and zero otherwise; *Education* is a rank variable denoting the education level of household head, with one for elementary or lower, two for high school, and three for college and above; *Age* denotes the age of household heads; and *Unemp* is a dummy variable set to one if a household head is unemployed, and zero otherwise. The third group of control variables contains information about the household financial condition, including household monthly income and net wealth. *Income* is the natural logarithm of one plus the value of monthly household income. *Net wealth* is the household wealth, including net equity of vehicle, business, liquid assets, and other assets minus total unsecured debt (Célerier and Matray 2019). We exclude home equity from wealth to mitigate the impact of housing.

The last group of control variables is state-related variables, including population and GDP growth. Specifically, *Log(pop)* is the natural logarithm of the population in the state in which a household worked. *GDP growth* is the annual GDP growth rate in the state. Moreover, following Célerier and Matray (2019), we adjust all the value-related variables (nominal prices) using the CPI in 2000. Finally, to mitigate the impact of outliers, we winsorize all continuous variables at the first and 99th percentiles. Table B.1 in Appendix B lists the detailed definitions of all the variables used in this study.

2.5. Descriptive Statistics

Table 1 reports descriptive statistics for the key variables used in our analysis. In our sample, *Log(1 + mdebt)* averages 4.780. We also find that the value of the first mortgage loan at origination (*Mdebt*) averages \$41,213, which is comparable to and lower than the \$53,685 shown in Chetty et al. (2017). The lower value is reasonable because we focus on the first mortgage loan at origination, whereas Chetty et al. (2017) report all mortgage debt. Further, we adjust the value by using the CPI in 2000, whereas Chetty et al. (2017) do not make an adjustment. In addition, *Dum_mdebt* has a mean value of 0.424, indicating that almost half of the households in our sample have mortgage loans. As for our key independent variable, we find that the mean value of *Local aggl* is 1.805, which is close to the average value of 1.50 found by Addoum et al. (2024), although in a different period of analysis.

Regarding our control variables, we show that on average, households have about three persons, including one child. In addition, 57.8% of household heads are

Table 1. Summary Statistics ($N = 41,366$)

Variables	Mean	Standard deviation	P25	Median	P75
$\text{Log}(1 + \text{mdebt})$	4.780	5.590	0	0	11.211
<i>Mdebt</i>	41,212.847	61,265.976	0	0	73,906.430
<i>Dum_mdebt</i>	0.424	0.494	0	0	1.000
<i>Local aggl</i>	1.805	2.252	0.786	1.157	1.812
<i>Size</i>	3.037	1.474	2.000	3.000	4.000
<i>Num_kid</i>	1.039	1.198	0	1.000	2.000
<i>Married</i>	0.578	0.494	0	1.000	1.000
<i>Female</i>	0.364	0.481	0	0	1.000
<i>Education</i>	2.547	0.621	2.000	3.000	3.000
<i>Age</i>	40.135	10.340	32.000	39.000	48.000
<i>Unemp</i>	0.058	0.233	0	0	0
<i>Income</i>	7.934	0.873	7.379	8.025	8.570
<i>Net wealth</i> (in thousands)	35.304	96.473	0.170	4.232	24.701
$\text{Log}(\text{pop})$	16.155	0.805	15.582	16.250	16.738
<i>GDP growth</i>	0.062	0.033	0.038	0.058	0.086

Notes. Table 1 reports descriptive statistics of the main variables used in our analysis. The sample comprises 41,366 household observations over the period 1984–2003. Variable definitions are provided in Appendix B.

married, and 36.4% of household heads are female. The household heads in our sample have an average (median) age of 40 (39), which is younger than the age documented by Chetty et al. (2017) and Célerier and Matray (2019). The reason could be that we restrict our sample to households with heads younger than 65. Meanwhile, during our sample period, about 5.8% of household heads are unemployed. The average logarithmic monthly income of households is 7.934, with an average monthly income of \$3,899, and the mean of household net wealth (in thousands) is 35.304; these values are comparable to those reported by Hsu et al. (2018) and Célerier and Matray (2019). Finally, the state-related variables show that the average state population is about 13.7 million, and the annual GDP growth averages about 6.2%, with a median value of 5.8%.

3. Empirical Analysis

3.1. Baseline Specification

To empirically examine the effect of local agglomeration on mortgage debt, we conduct the household-level analysis by using the following multivariate regression model:

$$\begin{aligned}
 \text{Mortgage Loan}_{ijmt} = & \alpha_0 + \alpha_1 \text{Local aggl}_{ijmt} + \alpha_2 \text{Size}_{ijmt} \\
 & + \alpha_3 \text{Num_kid}_{ijmt} + \alpha_4 \text{Married}_{ijmt} \\
 & + \alpha_5 \text{Female}_{ijmt} + \alpha_6 \text{Education}_{ijmt} \\
 & + \alpha_7 \text{Age}_{ijmt} + \alpha_8 \text{Unemp}_{ijmt} \\
 & + \alpha_9 \text{Income}_{ijmt} + \alpha_{10} \text{Net wealth}_{ijmt} \\
 & + \alpha_{11} \text{Log}(\text{pop})_{st} + \alpha_{12} \text{GDP growth}_{st} \\
 & + \theta_s + \varphi_t + \varepsilon_{ijmt}, \quad (1)
 \end{aligned}$$

where $\text{Mortgage Loan}_{ijmt}$ is a mortgage-related variable for household i , working in industry j and residing in MSA m in year t , which is proxied by $\text{Log}(1 + \text{mdebt})$ or

Dum_mdebt . The key independent variable is Local aggl_{ijmt} , which is defined in Section 2.3. Control variables include household-level and state-level variables, the definitions of which are discussed in Section 2.4. α_1 measures the effect of agglomerated economies on household mortgage debt. Following Addoum et al. (2024), we control for state fixed effect θ_s and year fixed effect φ_t in the model. Standard errors are clustered at the state level.

3.2. Main Results

Table 2 presents the results of estimating Equation (1). In column (1), the dependent variable is the natural logarithm of one plus mortgage loan amount at origination ($\text{Log}(1 + \text{mdebt})$), whereas in column (2), the dependent variable is an indicator variable (Dum_mdebt) denoting whether a household has a mortgage loan. We find, for both measures of mortgage loan, that the coefficients on local agglomeration are positive and statistically significant at the 1% level, suggesting that households working in locally agglomerated economies are more likely to have mortgage loans.¹³ This result is consistent with the career prospects view in which we argue that local agglomeration increases household career upward potential and provides downside protection because of benefiting from the knowledge spillover and a dense pool of job opportunities. In terms of economic significance, in column (1), an increase of one standard deviation in local agglomeration is associated with a 2% increase in mortgage loans.¹⁴ Given that the mean value of mortgage loans is \$41,213 in our sample, this increase represents an \$824 increase in mortgage loans. This result is comparable to the economic significance shown by Barrot et al. (2022), who examine the impact of import competition on household debts, with economic significance values of 2% and \$950, respectively. Column (2) of Table 2 shows that an increase of one

Table 2. Local Agglomeration and Mortgage Debt

Dependent variable	<i>Log(1+mdebt)</i> (1)	<i>Dum_mdebt</i> (2)
<i>Local aggl</i>	0.0430*** (2.92)	0.0043*** (3.24)
<i>Size</i>	−0.0778** (−2.26)	−0.0056* (−1.87)
<i>Num_kid</i>	0.6039*** (11.54)	0.0511*** (10.80)
<i>Married</i>	2.0035*** (19.70)	0.1739*** (18.32)
<i>Female</i>	−0.1256* (−1.93)	−0.0104* (−1.74)
<i>Education</i>	0.8726*** (15.57)	0.0725*** (14.20)
<i>Age</i>	0.0466*** (7.46)	0.0047*** (8.91)
<i>Unemp</i>	−0.1911** (−2.28)	−0.0167** (−2.33)
<i>Income</i>	1.6522*** (28.06)	0.1398*** (29.20)
<i>Net wealth</i>	0.0022*** (4.40)	0.0002*** (3.66)
<i>Log(pop)</i>	−0.0629 (−0.67)	−0.0040 (−0.46)
GDP growth	0.8489 (0.56)	0.1414 (1.03)
State FE	Yes	Yes
Year FE	Yes	Yes
Observations	41,366	41,366
Adjusted <i>R</i> ²	0.2235	0.2174

Notes. Table 2 reports the regression results of the effect of local agglomeration on household mortgage loans. *Local aggl* is described in Section 2. *Log(1+mdebt)* in column (1) is the natural logarithm value of mortgage loan at origination, whereas *Dum_mdebt* in column (2) is a dummy variable indicating if a household has a mortgage loan. Definitions of all the other variables are reported in Appendix B. Robust *t*-statistics clustered at the state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

standard deviation in local agglomeration is associated with a 1% increase in the probability of having mortgage debt, a 2% increase relative to the mean of *Dum_mdebt*.

With regard to control variables, we find that households with more family members have lower levels of mortgage loans and are less likely to take mortgage loans. In contrast, households with more children have more mortgage loans.¹⁵ This finding is consistent with the results reported by Ling and McGill (1998), who also report a positive impact of number of children on household debt. In addition, households with female heads are less likely to have mortgage loans (Mian and Sufi 2011). Moreover, we find that mortgage loan amount is positively associated with household heads' education level and age. This finding is in line with our expectation because households that are well educated and older are more likely to be able to afford a mortgage. Notably, unemployment is negatively associated with mortgage

loans. In addition, we find that household income and net wealth are positively related to mortgage loans. This result suggests that the role of local agglomeration extends beyond the wealth effect because we continue to observe a significant impact of local agglomeration after we control for wealth and income variables. Overall, the results in Table 2 suggest that local agglomeration significantly increases household mortgage loans, both their level and the probability of having them.

3.3. Robustness Check

3.3.1. Extended Sample. First, to enhance the relevance of our studies to the current mortgage market, we employ two different methods to expand the sample period. First, we use the HMDA data set to extend our sample period from 1990 to 2022. The biggest advantage of HMDA is that it provides mortgage application records, especially for mortgages that are not originated. By using HMDA data, we construct two dependent variables at the MSA-year level: the natural logarithm of the approved value of mortgage loans and the number of approved mortgage loans. In particular, to mitigate the concern that the number of mortgage loan approvals is driven by local population, we scale the number of approved mortgage loans by MSA population. In addition, we control for applicant-related characteristics, including loan-to-income ratio, applicants' income, applicants' gender, applicants' race, and loan types. We also control for the impact of unemployment rate and GDP growth rate, as well as local population if the dependent variable is the approved dollar amount. Consistent with our main results in Section 3.2, in columns (1) and (2) of panel A of Table 3, we find that local agglomeration significantly increases the approved amount and the approved number of mortgage loans.¹⁶ Second, we reconstruct our local agglomeration measure at the state-industry-year level, compared with the MSA-industry-year level constructed in the main tables. This change enables us to extend our sample to 2011 because SIPP provides the information of the state until 2008 panel. In columns (3) and (4), we find that local agglomeration is positively associated with the level of mortgage loans at origination and that households in agglomerated economies are more likely to have mortgages.¹⁷ The preceding results indicate that our main inference for the positive link between local agglomeration and mortgage loans also works for the recent sample period.

3.3.2. Alternative Specifications. To mitigate the concern that our results are sensitive to heterogeneity across households and local labor markets, and to validate the stability of our baseline results, we examine the validity of the local agglomeration effect by imposing different fixed effects. First, we replace the state and year fixed effects in our baseline regressions with state-by-year

Table 3. Robustness Check

Panel A: Extended sample						
Dependent variable	HMDA data:1990–2022		State-level measure:1984–2011			
	<i>Amount_appr</i> (1)	<i>No. appr</i> (2)	<i>Log(1+mdebt)</i> (3)	<i>Dum_mdebt</i> (4)		
<i>Local aggl</i>	0.6900*** (3.50)	0.0104*** (2.90)	0.0516** (2.10)	0.0049** (2.12)		
<i>Loan to income</i>	0.3744*** (8.05)	0.0026*** (3.03)				
<i>Male ratio</i>	0.8718 (1.04)	0.0159* (1.96)				
<i>White ratio</i>	−0.5524*** (−2.64)	−0.0086** (−2.56)				
<i>Conventional ratio</i>	1.6024*** (6.59)	0.0297*** (8.42)				
<i>Income average</i>	0.3500*** (3.66)	−0.0001 (−0.14)				
<i>Unemp rate</i>	−0.0536*** (−7.53)	−0.0012*** (−9.98)				
<i>GDP growth</i>	−0.1211 (−0.71)	0.0046 (1.14)				
<i>Log(pop)</i>	1.0734*** (7.90)					
Controls			Yes	Yes		
MSA FE	Yes	Yes	No	No		
Year FE	Yes	Yes	Yes	Yes		
State FE	No	No	Yes	Yes		
Observations	4,965	4,965	84,234	84,234		
Adjusted R ²	0.9605	0.8747	0.1992	0.1936		
Panel B: Alternative specifications						
Dependent variable	<i>Log(1+mdebt)</i> (1)	<i>Dum_mdebt</i> (2)	<i>Log(1+mdebt)</i> (3)	<i>Dum_mdebt</i> (4)	<i>Log(1+mdebt)</i> (5)	<i>Dum_mdebt</i> (6)
<i>Local aggl</i>	0.0433*** (2.96)	0.0043*** (3.28)	0.0325* (1.94)	0.0030** (2.04)	0.0357*** (2.72)	0.0037*** (3.24)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	No	No	No	No
MSA × Year FE	No	No	Yes	Yes	No	No
Occup × Year FE	No	No	No	No	Yes	Yes
Observations	41,366	41,366	41,366	41,366	41,366	41,366
Adjusted R ²	0.2263	0.2204	0.2295	0.2249	0.2357	0.2300

Notes. Panel A utilizes the extended sample. In the first two columns, we conduct an analysis at the MSA-year level, whereas the analysis in the last two columns is conducted at the individual level. Panel B tests the robustness of the baseline results by using alternative specifications. *Local aggl* is described in Section 2. Definitions of variables are reported in Appendix B. Robust *t*-statistics clustered at the MSA (for columns (1) and (2)) or state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

fixed effects, reported in columns (1) and (2) of panel B of Table 3. The objective of this alternative specification is to control for the time-specific sources of variation across states. Next, we instead control for MSA-by-year fixed effects in columns (3) and (4). The reason is that, within the same state, there could be time-varying MSA characteristics that affect household decisions on mortgage loans. As an example, within the state of New York, the house prices in Syracuse, New York, are significantly lower than in New York, New York. Consequently, laborers in Syracuse, New York, may be more likely than New Yorkers to have mortgages and to be

able to afford them. After imposing the state-by-year and MSA-by-year fixed effects, we find that the coefficients of local agglomeration still load positively.

Finally, it is possible that the heterogeneity across occupations drives our results. Residents who are chief executives are likely to have relatively higher wages than operators, which subsequently affects their decisions on mortgages. So, in the last two columns of panel B of Table 3, we use the occupation-by-year fixed effects to account for the time-varying differences across occupation groups. With the occupation-by-year fixed effects, we continue to report a positive and significant

effect of local agglomeration on mortgage loans. Overall, the stability of our results with different sets of fixed effects dispels the concern that our primary results are driven by omitted time-varying variables at the state, MSA, or occupation levels.

3.3.3. Other Robustness Tests. We further explore the sensitivity of our core evidence by conducting a number of robustness checks that are not tabulated but are available upon request. First, we examine whether our findings are driven by a few extremely large or small MSAs based on the aggregate labor supply. Second, we rule out the possibility that key industries drive our results by excluding the top 10% industries in each MSA. Finally, it is possible that our measure of local agglomeration actually proxies for other industry-level labor market characteristics that vary across geographies; we thus control for the impact of industry-related factors, including local industry concentration and innovation. In all the tests, we find that the positive impact of locally agglomerated economies on household mortgage loans remains large and statistically significant.

4. Endogeneity Tests

So far, our results show a strong positive association between local agglomeration and household mortgage debt. In this section, we examine the causal relation between these two factors. Our empirical findings may suffer from endogeneity threats stemming from reverse causality or omitted variables. For reverse causality, easier access to mortgage loans (and a higher level of mortgage supply) in certain areas could attract households to flow into those areas, leading to local agglomeration. Therefore, mortgage debt would lead to local agglomeration, not the other way around. In addition, if omitted variables increase both local agglomeration and household mortgage debt, then the link we observe could be spurious. In particular, institutional and economic conditions could simultaneously affect both local agglomeration and household mortgage debt.

4.1. IPO Activity as an Exogenous Shock

In this section, we use IPO activity as an exogenous shock to local agglomeration and then validate our findings under a DiD research design. Previous studies show that IPO activities in a locality are associated with an increase in labor employment (Borisov et al. 2021, Barrios et al. 2023). Specifically, Borisov et al. (2021) find that IPO firms increase employment by particularly large amounts in the one to two years following the offering. This increase occurs because of the improved ability to undertake acquisitions and the strategic shift toward commercialization after going public. In addition, an IPO offers an exit opportunity for entrepreneurial founders and employees to sell their shares and

create new and younger firms to pursue fresh entrepreneurial endeavors, subsequently leading to employment increases and raising local agglomeration (Babina et al. 2024). As going public is very likely to be exogenous to household mortgage debt and is closely related with local agglomeration, we use local IPO activity as an exogenous shock to local agglomeration and examine its impact on household mortgage debt.¹⁸ Following the research design of Barrios et al. (2023), we use the following model specifications to conduct our DiD analysis:

$$\begin{aligned} \text{Mortgage loan}_{ist} = & \beta_0 + \beta_1 \text{Post IPO}_{st} + \beta_2 \text{Size}_{ist} \\ & + \beta_3 \text{Num_kid}_{ist} + \beta_4 \text{Married}_{ist} \\ & + \beta_5 \text{Female}_{ist} + \beta_6 \text{Education}_{ist} \\ & + \beta_7 \text{Age}_{ist} + \beta_8 \text{Unemp}_{ist} + \beta_9 \text{Income}_{ist} \\ & + \beta_{10} \text{Net wealth}_{ist} + \beta_{11} \text{Log(pop)}_{st} \\ & + \beta_{12} \text{GDP growth}_{st} + \theta_s + \varphi_t + \varepsilon_{ist}, \end{aligned} \quad (2)$$

where $\text{Mortgage loan}_{ist}$ is proxied by $\text{Log}(1+mdebt)_{ist}$ or Dum_mdebt_{ist} of household i , residing in state s and year t . We obtain the IPO data from Thomson Financial's New Issues Database. Following prior literature, we exclude American depositary receipts (ADRs), real estate investment trusts (REITs), unit offerings, and IPOs with an offer price of less than five dollars (Ritter and Welch 2002). To define *Post IPO*, we first calculate the ratio of number of IPOs relative to the number of publicly listed firms for a given state-year. Next, if the ratio is in the top decile, we classify the year as an intensive IPO year. Subsequently, we define *Post IPO* as one after the state experiences the first intensive IPO activity, and zero otherwise. We calculate IPO activity at the state-year level because the MSA information is not available for IPO data. Our coefficient of interest is β_1 , which compares the change in mortgage loans for households experiencing intensive IPO activities versus those without intensive IPO activities. As the IPO activities will increase local agglomeration, we expect β_1 to be positive.

We report the DiD results in Table 4. As expected, in panel A, we find that the coefficient on *Post IPO* is positive and significant for both $\text{Log}(1+mdebt)$ and Dum_mdebt as dependent variables. This finding suggests that the intensive IPO activities significantly increase mortgage loans of households residing in the same state. As the IPO activity in our setting is proxied for local agglomeration, it supports our findings for the positive link between local agglomeration and household mortgage debt. At the same time, we admit that IPO activity could be associated with a wealth effect through which household mortgage debt is affected. In other words, the wealth of founders and employees might largely increase after a firm goes public, and increases in

Table 4. DiD Design: Using IPO Activity as a Shock

Panel A: Basic results from the DiD research design						
Dependent variable	$Log(1+mdebt)$ (1)		Dum_mdebt (2)			
<i>Post IPO</i>	0.2205** (2.03)		0.0196** (2.06)			
Controls	Yes		Yes			
State FE	Yes		Yes			
Year FE	Yes		Yes			
Observations	34,265		34,265			
Adjusted R^2	0.1952		0.1878			
Panel B: Validity of the staggered DID						
Dependent variable	Stacked DID		Placebo test		Randomization analysis	
	$Log(1+mdebt)$ (1)	Dum_mdebt (2)	$Log(1+mdebt)$ (3)	Dum_mdebt (4)	$Log(1+mdebt)$ (5)	Dum_mdebt (6)
<i>Post IPO</i>	0.3767*** (2.74)	0.0333*** (2.71)	−0.0402 (−0.35)	−0.0048 (−0.48)	−0.0189 (−0.20)	−0.0025 (−0.31)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cohort × State FE	Yes	Yes	No	No	No	No
Cohort × Year FE	Yes	Yes	No	No	No	No
State FE	No	No	Yes	Yes	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Observations	73,815	73,815	34,265	34,265	41,366	41,366
Adjusted R^2	0.2007	0.1909	0.1951	0.1877	0.2232	0.2170

Notes. This table reports the results of using IPO activities as shocks to local agglomeration. Panel A presents the basic results from the DiD research design. *Post IPO* is an indicator variable equal to one after a state experiences intensive IPO activity, and zero otherwise. Panel B tests the validity of our staggered DiD research design. In the first two columns, we conduct a stacked DiD analysis. In the next two columns, we conduct the placebo test by making year $t - 3$ the pseudo-adoption year (year t is the actual IPO activity year). Finally, we randomly assign activity year to states. Definitions of all variables are reported in Appendix B. Robust t -statistics clustered at the state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

household mortgage loans could possibly be affected by a wealth effect rather than a local agglomeration effect. Controls of household wealth and income could mitigate the wealth effect to some extent. That is, we find that IPO activity significantly increases household mortgage debt even when we consider the impact of household income and net wealth.

In panel B of Table 4, we conduct additional tests in support of our DiD research design. First, we estimate our DiD coefficients using a stacked DiD (Cengiz et al. 2019, Baker et al. 2022). Baker et al. (2022) indicate that the estimates of the staggered DiD could be biased when treatment effect heterogeneity exists either across groups or over time because, in a staggered DiD, earlier-treated units act as controls for the later-treated units. To address this concern, we implement a stacked DiD analysis.¹⁹ We construct a “clean” cohort for each time-specific implementation. Each cohort is observed in a $[-5, 5]$ window (i.e., five years before the adoption year to five years after it).²⁰ These cohorts are then stacked together, and the regression is estimated on the stacked data set, with cohort-by-state and cohort-by-year fixed effects. Standard errors are clustered at the state-by-cohort level. In the first two columns of panel B, we find

that the coefficients on *Post IPO* remain positive and significant, validating our basic DiD results. In the next two columns, we conduct a placebo test by setting year $t - 3$ as the pseudo-shock year (year t is the actual shock year) and reestimate the coefficients using the pseudo-*Post IPO*. We find that, when we move the event year back by three years, the coefficient on *Post IPO* is no longer significant. Next, we randomly assign IPO year to states in columns (5) and (6).²¹ *Post IPO* is equal to one for states after the assigned IPO years and equal to zero before the assigned IPO years or for firms in the assigned control states. If the documented link between IPO activity and household mortgage debt is driven by omitted variables in the regression, we would also observe a statistically significant coefficient on *Post IPO*. However, the coefficients on *Post IPO* in the last two columns are not significant. In sum, the results in Table 4 validate our staggered research design as well as the link between local agglomeration and household mortgage loans.

4.2. Instrumental Variable Analysis

We further tackle the reverse causality issue and omitted variable concerns in an instrumental variable (IV) framework. In particular, we instrument local agglomeration

by using the industry-level exposure to the United States granting PNTRs to China in 2001. The U.S. PNTRs reduce China's uncertainty in a favorable trade partnership with the United States. Such reduction in uncertainty largely boosts the imports from Chinese firms to the United States, subsequently posing a threat to U.S. firms and reducing the labor supply of U.S. firms. As background, China joined the World Trade Organization (WTO) in December 2001. Before joining the WTO, China had enjoyed MFN status as a trading partner of the United States since 1980, which means that China-made goods have a lower tariff rate when exported to the United States. However, China's MFN status is updated every year, and American manufacturers often lobby against it (Pierce and Schott 2016). In other words, China's MFN status is subject to political uncertainty.

As the United States's biggest trading partner, China removed its uncertainty associated with favorable tariffs when it joined the WTO, leading to a significant increase in Chinese firms' investment and import competition with U.S. firms (Pierce and Schott 2016). As indicated by Pierce and Schott (2016), U.S. industries exposed to increased Chinese competition experience significant employment losses. Generally, the PNTRs have a relatively large impact on the trade industry and less impact on other industries. Thus, we implement our IV analysis by including a tradable sector indicator. More precisely, we use the interaction term of the tradable sector indicator with post-PNTRs as our IV. To define tradable sector, we include the following 10 broad sectors by referring to prior literature (see, for instance, Mian and Sufi 2014): agriculture, forestry, and fishing; mining; construction; manufacturing; transportation, communications, electric, gas, and sanitary services; wholesale trade; retail trade; finance, insurance, and real estate; services; and public administration.

Overall, the PNTRs satisfy the criteria of relevance well because they are closely related to the labor supply of U.S. trade firms. To satisfy the exclusion restriction, PNTRs should affect mortgage loans only via the channel of local agglomeration. However, PNTRs could possibly affect the price of consumption goods. If so, the income effect from consumer market rather than labor market could be the underlying channel through which PNTRs affect household mortgage debt. We address the potential violation of the exclusion restriction caused by the price and income effect in two ways. First, given that we have adjusted all the value-related variables, including mortgage loans and household income, by using the CPI in 2000, the concern about the price effect can be alleviated to some extent. Second, we control for income square and even industry income in the regression. In particular, we include a squared income control in addition to the standard income control. Moreover, we compute and include annual income averaged over the pre- and post-PNTRs samples for each industry. This control

variable is aimed at capturing the change of industry income as a result of the shock of PNTRs.

In Table 5, we show the results of the IV analysis. In columns (1)–(3), we show the results for the full sample, and in columns (4)–(6), we report the results in the seven-year event window to mitigate the concern of unbalanced pre- versus postperiod observations. Specifically, columns (1) and (4) report the results of first-stage regression, and columns (2) and (5) and columns (3) and (6) show the second-stage results of $\text{Log}(1+mdebt)$ and Dum_mdebt , respectively. In the first-stage regression, the variable of interest is the trade sector indicator interacted with the post-PNTRs period, namely, $\text{Trade_sector} \times \text{Post_PNTR}$. We also include the trade sector indicator (Trade_sector) in the regression and exclude the term of Post_PNTR to address the concern of multicollinearity with the year fixed effect. As expected, in both samples, the coefficients of $\text{Trade_sector} \times \text{Post_PNTR}$ load negatively and significantly. That means, since China joined the WTO in 2001, Chinese firms pose a great threat to local U.S. firms, negatively affecting agglomerative patterns in America. In addition, the Kleibergen-Paap Wald F -statistic is 24.857, which is significantly larger than the critical value (19.93), rejecting the null hypothesis of a weak instrument. As for the second-stage regressions, we continue to find a positive and statistically significant coefficient on our key independent variable. In sum, our IV analysis validates our baseline regression and suggests that local agglomeration has a positive effect on household mortgage debt.

5. Channel Tests

In this section, we examine the underlying channels that drive the increased mortgage loans in agglomerated economies. First, we conduct demand-side analysis to examine the impact of local agglomeration on housing demand and mortgage demand. Second, we conduct supply-side analysis to examine the impact of local agglomeration on mortgage supply.²² Finally, from the perspective of loan contracts, we examine whether the increase in mortgage loans is because of purchasing larger houses or because of smaller down payments.

From demand side, agglomerated economies could significantly affect housing and mortgage demand. This outcome is because households working in agglomerated economies have higher expectations for the future, leading to a more assertive stance in the housing market. To capture housing demand, we follow prior literature and use three measures: the dollar amount of mortgage loans in applications and the amount of approved loans (Agarwal et al. 2022) and the house value deflated by hedonic-adjusted price index (Han 2010). Following Barrot et al. (2022), we aggregate mortgage loans in applications (and in approvals) to the MSA-year level and proceed with our analysis at that

Table 5. Instrumental Variable Analysis

Dependent variable	Full sample			Seven-year window sample		
	First stage	Second stage		First stage	Second stage	
	<i>Local aggl</i> (1)	<i>Log(1+mdebt)</i> (2)	<i>Dum_mdebt</i> (3)	<i>Local aggl</i> (4)	<i>Log(1+mdebt)</i> (5)	<i>Dum_mdebt</i> (6)
<i>Trade_sector</i> × <i>Post_PNTR</i>	−0.2759* (−1.72)			−0.4071* (−1.95)		
<i>Trade_sector</i>	1.7533*** (5.45)			1.9350*** (6.69)		
<i>Local aggl</i>		0.1674** (2.14)	0.0158** (2.22)		0.1644** (2.26)	0.0153** (2.39)
<i>Income square</i>	−0.0030 (−0.34)	−0.0642** (−2.34)	−0.0089*** (−3.72)	0.0156 (0.99)	0.1067*** (3.82)	0.0029 (1.20)
<i>Industry income</i>	0.0245*** (2.64)	0.0349*** (4.47)	0.0030*** (4.32)	0.0194** (2.13)	0.0239*** (2.94)	0.0020*** (2.83)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	26,169	26,169	26,169	13,754	13,754	13,754
Adjusted R ²	0.1598	0.1615	0.1529	0.1645	0.1522	0.1435

Notes. This table reports IV regression estimates. *Local aggl* is described in Section 2 and is instrumented by increased competition for tradable sectors following the United States granting PNTRs to China. Tradable sectors include the following 10 broad sectors: agriculture, forestry, and fishing; mining; construction; manufacturing; transportation, communications, electric, gas, and sanitary services; wholesale trade; retail trade; finance, insurance, and real estate; services; and public administration. Definitions of other variables are reported in Appendix B. Robust *t*-statistics clustered at the state level are reported in parentheses beneath each estimate. Kleibergen-Paap rk Wald *F*-statistic: 24.857 (for seven-year window sample).

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

level. Specifically, we define *Amount_appl* and *Amount_appr* as the natural logarithm of the aggregate value of mortgage loans in applications and approvals. *Hedonic-adjusted house value* is calculated as the property value adjusted by hedonic-adjusted price index. Following Han (2010), we select the price index of Madison, Wisconsin, in 1984 as the benchmark to calculate hedonic-adjusted price index.²³ The analysis on *Hedonic-adjusted house value* is done at the individual level. We present the results in columns (1)–(3) in panel A of Table 6. We find that local agglomeration is significantly and positively associated with housing demand for those proxies. Regarding the mortgage demand, we also conduct the analysis at the MSA-year level and use the number of mortgage loan applications scaled by MSA population to capture mortgage demand (Allen et al. 2023). The results in column (4) of Panel A in Table 6 show that local agglomeration has a significant and positive effect on mortgage demand.

Next, we turn to discuss the impact of supply-side factors. Specifically, we use the mortgage approval rate and the number of granted loans to measure the supply-side driver. *Approval rate* is defined as the ratio of the number of approved loans to the number of loan applications. *No. appl* is calculated as the number of granted loans in each MSA scaled by MSA population. We present the supply-side analysis in panel B of Table 6. The first two columns of panel B show that the approval rate and the number of approved mortgage loans

significantly increase with agglomerated economies. This finding indicates that loan officers are more likely to approve loan applications from MSAs with more agglomerated economies. In addition, to alleviate the concern that banks in clustered areas are compelled to lend to workers, in columns (3) and (4), we remove from our sample large labor markets in which local agglomeration is in the top 10%.²⁴ We find that the positive coefficient on local agglomeration continues to be significant.

Finally, upon observing the rise in mortgage loans, we differentiate the effect of smaller down payment from the effect of purchasing a large house. Particularly, we decompose our dependent variable (amount of mortgage loan at origination) into two parts: loan-to-value (*LTV*) and property value. Specifically, the amount of mortgage loan at origination = (mortgage loan at origination)/(property value at origination) × property value at origination, where (mortgage loan at origination)/(property value at origination) is *LTV* (Agarwal et al. 2017). A larger *LTV* indicates a smaller down payment, and a greater property value corresponds to a larger house being purchased. Then, we examine how local agglomeration affects *LTV* and original property value, respectively. We present the results in panel C of Table 6. In column (1), we find that local agglomeration is positively related to the property value at the time of purchase. Meanwhile, in column (2), we find that the coefficient on local agglomeration loads positively at the 1% level when the dependent variable is *LTV*. These findings suggest that a higher mortgage debt

Table 6. Channel Test

Panel A: Demand-side analysis				
Dependent variable	Housing demand			Mortgage demand
	<i>Amount_appl</i> (1)	<i>Amount_appr</i> (2)	<i>Hedonic-adjusted house value</i> (3)	<i>No. appl</i> (4)
<i>Local aggl</i>	0.8867*** (4.45)	0.8142*** (4.16)	0.2244*** (6.54)	0.0118*** (3.04)
<i>Loan to income</i>	0.3554*** (7.34)	0.2495*** (4.05)		0.0019** (2.54)
<i>Male ratio</i>	−0.3271 (−0.30)	0.2009 (0.18)		0.0092 (0.80)
<i>White ratio</i>	−0.8791* (−1.94)	−0.7556* (−1.69)		−0.0200*** (−3.00)
<i>Conventional ratio</i>	1.7335*** (4.58)	1.5634*** (4.04)		0.0448*** (9.64)
<i>Income average</i>	0.1034 (1.00)	0.1528 (1.43)		−0.0047*** (−4.04)
<i>Unemp rate</i>	−0.0206*** (−3.05)	−0.0258*** (−3.64)		−0.0015*** (−8.58)
<i>GDP growth</i>	−0.6233*** (−3.17)	−0.5336*** (−2.72)		−0.0098 (−1.41)
<i>Log(pop)</i>	0.2626** (2.07)	0.7519*** (6.79)		
Controls			Yes	
MSA FE	Yes	Yes	No	Yes
State FE	No	No	Yes	No
Year FE	Yes	Yes	Yes	Yes
Observations	2,777	2,777	20,711	2,777
Adjusted <i>R</i> ²	0.9756	0.9729	0.2594	0.9124
Panel B: Supply-side analysis				
Dependent variable	Mortgage supply		Mortgage supply (excluding MSAs with top 10% local agglomeration)	
	<i>Approval rate</i> (1)	<i>No. appr</i> (2)	<i>Approval rate</i> (3)	<i>No. appr</i> (4)
<i>Local aggl</i>	0.0634*** (3.64)	0.0107*** (3.64)	0.0664*** (3.33)	0.0116*** (4.05)
<i>Loan to income</i>	−0.0120** (−2.29)	0.0008 (1.55)	−0.0127** (−2.17)	0.0009 (1.55)
<i>Male ratio</i>	0.4712*** (4.30)	0.0175** (2.03)	0.4642*** (3.92)	0.0163* (1.79)
<i>White ratio</i>	−0.0087 (−0.16)	−0.0117*** (−2.62)	−0.0053 (−0.10)	−0.0133*** (−2.72)
<i>Conventional ratio</i>	−0.0229 (−0.87)	0.0279*** (8.31)	−0.0234 (−0.83)	0.0264*** (7.79)
<i>Income average</i>	0.0337*** (4.54)	−0.0017** (−2.18)	0.0377*** (4.66)	−0.0018** (−2.10)
<i>Unemp rate</i>	−0.0024** (−2.46)	−0.0013*** (−9.14)	−0.0022** (−2.11)	−0.0013*** (−9.37)
<i>GDP growth</i>	0.0913*** (2.92)	−0.0040 (−0.85)	0.0927*** (2.79)	−0.0039 (−0.82)
<i>Log(pop)</i>	0.0056 (0.29)		0.0013 (0.07)	
MSA FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2,777	2,777	2,498	2,498
Adjusted <i>R</i> ²	0.8736	0.8994	0.8727	0.9004
Panel C: Big house purchase and small down payment				
Dependent variable	Big house <i>Property value</i>		Down payment <i>LTV</i>	
	(1)		(2)	
<i>Local aggl</i>	0.0503*** (2.86)		0.0048*** (4.93)	

Table 6. (Continued)

Panel C: Big house purchase and small down payment		
Dependent variable	Big house Property value (1)	Down payment LTV (2)
Controls	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Observations	41,366	41,366
Adjusted R ²	0.3013	0.1769

Notes. This table examines the channels that drive the increase in mortgage loans. In panel A, we examine the demand effect, manifested in housing demand and mortgage demand. For columns (1), (2), and (4), we conduct analysis at the MSA-year level. Column (3) is analyzed at the individual level. In panel B, we examine the effect of local agglomeration on mortgage supply at the MSA-year level. In panel C, at the individual level, we examine whether the increase in mortgage loans is driven by larger housing purchasing or smaller down payment. For local agglomeration, we multiple it by 100 except when the dependent variable is hedonic-adjusted house value for interpretation convenience. Definitions of all the other variables are reported in Appendix B. Robust *t*-statistics clustered at the MSA level or state level (for column (3) of panels A and C) are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

for households in agglomerated economies could be driven by households’ willingness to buy a larger house and simultaneously driven by a lower down payment.

6. Mechanism Tests

In this section, we examine the underlying mechanisms behind the career prospects view. We argue that career prospects play a role through the risk-based effect. Such a risk-based effect is evident in upward potential and downside protection in the labor market. To validate this argument, we examine whether the role of local agglomeration varies with individuals’ working skills and employment status.

6.1. Upside Potential: Working Skills

Regarding upside potential, local agglomeration increases the opportunities for labor to acquire better skills. Specifically, agglomerated economies facilitate the transmission of technologies and knowledge, resulting in better career prospects (Marshall 1890, Addoum et al. 2024). Thus, we expect that the role of local agglomeration is stronger for skilled employees.

Panel A of Table 7 reports results regarding labor skills. Following Acemoglu and Autor (2011) and Addoum et al. (2024), we classify workers as skilled if their occupation is managerial, professional, or technical. As expected, the positive coefficients on *Local aggl* × *Skill* in columns (1) and (2) indicate that skilled employees are more likely to get mortgage loans if they work in agglomerated economies. This result supports the upside potential role of local agglomeration in which we argue that agglomerated local labor markets are likely to hold enhanced prospects for skilled employees. It is also consistent with the findings of Addoum et al. (2024), who show that local agglomeration increases human capital of laborers. Collectively, these results suggest that local agglomeration increases the career upside potential of skilled laborers and therefore increases their competitiveness in mortgage markets.

6.2. Downside Protection: Unemployment

As a further test of the risk-based effect of local agglomeration, we next examine whether local agglomeration provides protection to individuals, especially to those who are perceived to be risky. Agglomerated economies provide abundant employment opportunities for workers. Such a broad labor market improves employee position matching and reduces unemployment spells, thereby reducing the mortgage default risk. In particular, we interact local agglomeration with unemployment indicator, with the expectation that unemployed individuals are more likely to get mortgages if they are in more agglomerated economies than in less agglomerated economies. This outcome is because the thick labor market of local agglomeration acts as a protective factor for risky (unemployed) individuals.

We show the results in panel B of Table 7. *Unemp* is a dummy variable set to one if the household head is unemployed, and zero otherwise. In columns (1) and (2), we find that the coefficient on *Unemp* is significantly negative, whereas the coefficient on *Local aggl* × *Unemp* is significantly positive. This result indicates that local agglomeration weakens the negative association between unemployment and mortgage loans. Put differently, agglomerated economies provide downside-protection risk to individuals who are perceived to be risky.²⁵ Collectively, our finding is consistent with our argument of the protective role of local agglomeration, and it also supports the argument that labor market condition is an important factor in predicting household mortgage debt (Hsu et al. 2018).

7. Alternative Explanation

Although we have utilized DiD and IV analyses to mitigate the endogeneity concern, the concern of omitted variables cannot be excluded. For example, it is possible that households choose to live in agglomerated

Table 7. Mechanism Tests

Panel A: Career prospects mechanism: better skills		
Dependent variable	<i>Log(1+mdebt)</i> (1)	<i>Dum_mdebt</i> (2)
<i>Local aggl</i>	0.0327** (2.24)	0.0035** (2.68)
<i>Local aggl</i> × <i>Skill</i>	0.0386** (2.57)	0.0030** (2.13)
<i>Skill</i>	0.1467* (1.71)	0.0087 (1.18)
Controls	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Observations	41,366	41,366
Adjusted <i>R</i> ²	0.2237	0.2175
Panel B: Career prospects mechanism: unemployment		
Dependent variable	<i>Log(1+mdebt)</i> (1)	<i>Dum_mdebt</i> (2)
<i>Local aggl</i>	0.0373** (2.60)	0.0037*** (2.87)
<i>Local aggl</i> × <i>Unemp</i>	0.1059** (2.07)	0.0095* (2.00)
<i>Unemp</i>	−0.3758*** (−3.14)	−0.0333*** (−3.22)
Controls	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Observations	41,366	41,366
Adjusted <i>R</i> ²	0.2235	0.2175

Notes. This table examines the career prospects view by focusing on the role of skills (panel A) and unemployment (panel B). *Skill* is an indicator variable equal to one if the occupation of the labor head is managerial, professional, or technical, and zero otherwise. *Unemp* is a dummy variable set to one if the household head is unemployed, and zero otherwise. Definitions of all the other variables are reported in Appendix B. Robust *t*-statistics clustered at the state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

economies based on latent factors that may be correlated with household mortgage debt decisions (Mian and Sufi 2011, Hsu et al. 2018, Addoum et al. 2024). In this section, we consider several potential confounding factors.

7.1. Migration

First, migration may codetermine a household's location and its mortgage debt. Specifically, households may locate based on unobservable factors such as optimism about a city's economic prospects, which can be correlated with mortgage debt. Previous studies suggest that households' location decision is possibly endogenous. For example, Branikas et al. (2020) demonstrate that households sort into locations based on their positive views of local economic prospects. To mitigate such selection concerns, we examine the link between local agglomeration and migration.

We obtain the data of migration from the CPS. The CPS provides the information on an individual's last

year and current year residence at the state level. The working industry of each individual is also provided by the CPS. Based on this information, we use two measures to gauge migration: total inflow and net inflow of employees. Specifically, for each state-industry-year, we first calculate the total inflow of employees and total outflow of employees. Next, the net inflow is defined as the difference between the total inflow of employees and total outflow of employees. To match with the migration data for a given state-industry-year, we recalculate local agglomeration for each state-industry-year pair and then run the regression to examine the impact of local agglomeration on household migration. The result is reported in panel A of Table 8. We find that the coefficients on local agglomeration are not significant for either total inflow or net inflow of employees. This finding suggests that a household's location choice is not endogenously determined by local agglomeration, which mitigates the selection concern.²⁶

7.2. Housing Supply Elasticity and House Prices

In this section, we consider two important latent factors that may codetermine local agglomeration and

Table 8. Alternative Explanations

Panel A: Migration				
Dependent variable	<i>Net inflow</i> (1)		<i>Total inflow</i> (2)	
<i>Local aggl</i>	−0.2983 (−0.20)		0.1394 (0.07)	
State FE	Yes		Yes	
Year FE	Yes		Yes	
Observations	954		954	
Adjusted R ²	0.0972		0.2092	
Panel B: Housing supply and house prices				
Dependent variable	<i>Log(1+mdebt)</i> (1)	<i>Dum_mdebt</i> (2)	<i>Log(1+mdebt)</i> (3)	<i>Dum_mdebt</i> (4)
<i>Local aggl</i>	0.0346** (2.12)	0.0033** (2.31)	0.0349** (2.18)	0.0033** (2.35)
<i>Housing supply</i>	0.2711** (2.27)	0.0275** (2.37)	0.2727** (2.27)	0.0275** (2.37)
<i>Hprice growth</i>			−0.6569 (−0.68)	−0.0247 (−0.28)
Controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	38,968	38,968	38,968	38,968
Adjusted R ²	0.2269	0.2209	0.2269	0.2208

Notes. This table rules out alternative explanations. Panel A mitigates the impact of migration, and panel B rules out the impact of housing supply and house prices. *Migration* is defined as total inflow or net inflow of employees. *Housing supply* is measured by housing supply elasticity, and *Hprice growth* is defined as the annual growth rate of house prices in states. *Local aggl* is described in Section 2. Definitions of all the other variables are reported in Appendix B. Robust *t*-statistics clustered at the state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

mortgage choices, that is, housing supply and house prices. First, the housing supply is very likely to be particularly high in agglomerated areas because of the well-developed economies. The mechanism through which local agglomeration increases mortgage loans could then be the housing supply rather than the career prospects of employees. Second, previous studies find that house price is a key driver of household mortgage debt (Mian and Sufi 2011). For example, Mian and Sufi (2011) find that households increase their mortgage debt by extracting the home equity caused by increased house prices. As such, laborers may choose to live in agglomerated economies for the consideration of house prices.

To mitigate these concerns, we control for the impact of housing supply and house prices. Specifically, we add housing supply elasticity and house price growth in the regression. Housing supply elasticity is defined as the ratio of relative changes in the inventory of housing to relative changes in the price of housing. We obtain the data on housing supply elasticity for each MSA from Saiz (2010). House price growth is defined as the annual house price growth in states.²⁷ The results are reported in panel B of Table 8. We continue to observe a significant effect of local agglomeration on mortgage loans after we control for the impact of housing supply elasticity and house price growth. These results suggest that housing supply and house price appreciation do not absorb the effect of local agglomeration on mortgage debt. Therefore, the results exclude the alternative interpretation that our main results are driven by housing supply or house price appreciation in agglomerated economies.

In additional tests that are not tabulated but are available upon request, we also rule out the possibility that unemployment insurance or lower mortgage interest rates in agglomerated economies drive our results.

8. Further Analyses

In this section, we conduct further analyses. First, we examine the impact of local agglomeration on household income and wealth. We expect that laborers in agglomerated economies benefit from knowledge spillovers (Glaeser et al. 1992, Porter 1998) that lead to higher income and wealth. In addition, agglomerated economies also benefit employers in terms of productivity and profitability (Davis et al. 2014), which, in turn, increases laborers' income and wealth.²⁸ Such a wealth effect therefore increases the level of household mortgage, as well as the probability of having mortgage loans.

Panel A of Table 9 reports the results. As expected, in columns (1) and (2), we find that local agglomeration significantly increases household wealth and income. This finding supports the wealth effect of agglomerated economies. However, it is noticeable that local agglomeration

Table 9. Further Analysis

Panel A: Wealth effect				
Dependent variable	Log (1+Net wealth)		Income	
	(1)		(2)	
<i>Local aggl</i>	0.0075*		0.0094***	
	(1.78)		(4.26)	
Controls	Yes		Yes	
State FE	Yes		Yes	
Year FE	Yes		Yes	
Observations	35,843		41,366	
Adjusted R ²	0.4737		0.4701	
Panel B: Payment behaviors				
Dependent variable	Mortgage payment		Payment year	
	(1)		(2)	
<i>Local aggl</i>	0.0089		0.0984***	
	(0.94)		(2.91)	
Controls	Yes		Yes	
State FE	Yes		Yes	
Year FE	Yes		Yes	
Observations	28,756		41,296	
Adjusted R ²	0.2834		0.1897	
Panel C: Firm size effect				
Dependent variable	Log(1+mdebt)		Dum_mdebt	
	Large firm	Small firm	Large firm	Small firm
	(1)	(2)	(3)	(4)
<i>Local aggl</i>	0.0321*	0.0558***	0.0033**	0.0053***
	(2.01)	(3.59)	(2.31)	(3.70)
Controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	31,778	9,588	31,778	9,588
Adjusted R ²	0.2010	0.3033	0.1943	0.2979
Panel D: Industry variation				
Dependent variable	Log(1+mdebt)		Dum_mdebt	
	(1)		(2)	
<i>Local aggl</i> × <i>Manufacturing</i>	0.0763***		0.0073***	
	(5.89)		(6.05)	
<i>Local aggl</i> × <i>Wholesale</i>	0.0447		0.0037	
	(0.81)		(0.74)	
<i>Local aggl</i> × <i>Personal service</i>	−0.0826		−0.0076	
	(−1.24)		(−1.41)	
<i>Local aggl</i> × <i>Agrimining</i>	−0.0505		−0.0042	
	(−1.57)		(−1.59)	
<i>Local aggl</i> × <i>Other industries</i>	−0.0326		−0.0026	
	(−1.53)		(−1.35)	
Controls	Yes		Yes	
State FE	Yes		Yes	
Year FE	Yes		Yes	
Observations	41,366		41,366	
Adjusted R ²	0.2241		0.2181	

Notes. This table presents further analyses. Panel A examines the impact of local agglomeration on income and net wealth. Panel B examines the impact of local agglomeration on payment behaviors. Panel C examines the impact of local agglomeration in big and small firm groups. Panel D examines how the effect of local agglomeration on mortgage debt varies across industries. Definitions of all variables are reported in Appendix B. Robust *t*-statistics clustered at the state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

plays a role beyond the income and wealth effect. This result is because we continue to observe a positive and significant impact of local agglomeration after we control for household wealth and income in our main regression.

Second, as further evidence, we discuss the impact of local agglomeration on repayment behaviors. Although the amount of a mortgage loan at the time of origination is less affected by borrowers' repayment behaviors, we provide further analysis of payment behaviors to make our story more complete. We focus on monthly mortgage payment and payment years. In particular, *Mortgage payment* is the natural logarithm of one plus monthly mortgage debt payment. *Payment year* is the number of years over which payments are to be made. In panel B of Table 9, we find that local agglomeration lengthens the payment years of mortgage loans, whereas it has no significant impact on monthly mortgage payments.

Third, we examine the role of firm size on the relation between local agglomeration and household mortgage debt. We use the number of employees of firms to capture firm size. In the SIPP survey, households are asked the question, "About how many persons are employed by employer?" The answer is coded 1 if the number is under 25, 2 if the number is between 25 and 99, and 3 if the number is greater than 100. Utilizing this question, we define companies with more than 100 employees as large companies and companies with less than 100 employees as small companies. We present the results in panel C of Table 9. We find that the impact of local agglomeration on mortgage debt is significant for households working either for large companies or for small companies.²⁹ Overall, these results suggest that the impact of local agglomeration does not vary with firm size. In other words, local agglomeration plays a role even for households working for small firms in clustered areas. This finding further supports our argument that agglomerated economies act as protection for households that are thought to be risky (i.e., those working for small firms).

Finally, we investigate whether the importance of local agglomeration varies across industries. We categorize five key industries based on the leading industries presented in Appendix A. These key industries include manufacturing, wholesale, personal and entertainment service, agriculture and mining, and other industries. The results shown in panel D of Table 9 indicate that the impact of local agglomeration on mortgage debt is significant for the manufacturing industry, whereas the

other industries do not experience a significant impact from local agglomeration.³⁰

9. Conclusion

In this paper, we analyze the role of local agglomeration in household mortgage loans. By using the household survey, we document a strong and positive relationship between local agglomeration and household mortgage loans, including both mortgage amount and the probability of having mortgage loans. We further show that this pattern is significant and robust under a DiD research design and an IV framework. The demand-side and supply-side analyses show that the increase in mortgage loans is driven by increased housing and mortgage demand and by mortgage supply. In addition, the increased mortgage loans are because of households purchasing larger houses and households paying smaller down payments. Our mechanism test validates the risk-based effect of local agglomeration. Specifically, we find that the effect of local agglomeration is stronger for employees with greater upward potential (evident in better working skills) and local agglomeration provides downside protection by attenuating the negative link between mortgage loans and unemployment.

In a nutshell, our results validate the theoretical argument of agglomerated economies. That is, local agglomeration affects labor market conditions, which is manifested in knowledge spillover and a better employee-position matching process (Glaeser et al. 1992, Porter 1998, Duranton and Puga 2004). Such dynamic interactions increase laborers' career prospects, subsequently increasing the affordability and availability of mortgage loans for the laborers. Our study contributes to the literature on mortgage debt. In particular, our findings enlarge the determinant vector of mortgage debt by including location-based factors. Our results improve the understanding of mortgage debt, which has been proven to have significant consequences for the real economy. Our study also adds new evidence to studies of local agglomeration by extending its impact to household debts.

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Appendix A. Top 15 Local Agglomeration Economies

Table A.1. Top 15 Local Agglomeration Economies

Rank	MSA	Industry	Local agglomeration
1	Las Vegas, NV	Hotels and motels	14.350
2	Detroit, MI	Motor vehicles and motor vehicle equipment	11.763
3	Fresno, CA	Agricultural production, crops	10.104
4	Greensboro-Winston Salem, NC	Furniture and fixtures	8.256
5	Austin, TX	Computers and related equipment	7.124
6	Washington, DC/MD/VA	National security and international affairs	6.822
7	Albuquerque, NM	Research, development, and testing services	6.366
8	Syracuse, NY	Miscellaneous retail stores	6.053
9	Nashville, TN	Computers and related equipment	6.044
10	Albany-Schenectady-Troy, NY	Administration of human resources programs	5.530
11	Norfolk-Virginia Beach-Newport News, VA	National security and international affairs	5.375
12	Salt Lake City-Ogden, UT	National security and international affairs	5.352
13	Salt Lake City-Ogden, UT	Credit agencies	5.348
14	Dayton-Springfield, OH	National security and international affairs	5.313
15	Orlando, FL	Miscellaneous entertainment and recreation services	5.295

Notes. This table lists the 15 most locally agglomerated MSA-industry pairs in our sample, based on our local agglomeration measure described in Section 2. We focus on the 50 largest MSAs based on population and the 50 largest industries based on aggregate labor supply. We use the industry classification from the 1990 Census, and the MSA is the four-digit code used in the 1990 Census.

Appendix B. Variable Definitions

Table B.1. Variable Definitions

Variable	Definitions
<i>Local aggl</i>	The labor supply share of an industry in the local labor market scaled by the industry’s labor supply share across the country.
<i>Mdebt</i>	The principal amount of the first mortgage loan at origination.
<i>Log(1+mdebt)</i>	Natural logarithm of one plus the principal amount of first mortgage loan at origination.
<i>Dum_mdebt</i>	A dummy variable equal to one if a household has a mortgage loan, and zero otherwise.
<i>Size</i>	The number of individuals in a household.
<i>Num_kid</i>	The number of children in a household.
<i>Married</i>	A dummy variable equal to one if the household head is married, and zero otherwise.
<i>Female</i>	A dummy equal to one if the household head is female, and zero otherwise.
<i>Education</i>	A rank variable denoting the household head’s education level, with one for elementary or lower, two for high school, and three for college and above.
<i>Age</i>	The household head’s age.
<i>Unemp</i>	A dummy variable set to one if the household head is unemployed, and zero otherwise.
<i>Income</i>	Natural logarithm of one plus the value of monthly household income.
<i>Net wealth</i>	Household net wealth in thousands, which is the sum of net equity of vehicle, business, liquid assets, and other assets minus total unsecured debt. Unsecured debt includes credit cards, bank loans, and store or medical bills.
<i>Log(pop)</i>	Natural logarithm of the population in the state (or in the MSA for HMDA data) in which a household works.
<i>GDP growth</i>	The annual GDP growth rate in the state (or in the MSA for HMDA data).
<i>Amount_appr</i>	The natural logarithm of the value of approved mortgage loans.
<i>No. appl</i>	The number of mortgage loan applications in an MSA divided by MSA population.
<i>Loan to income</i>	The average ratio of the loan amount to the applicant’s income in the MSA.
<i>Male ratio</i>	The proportion of applications from male applicants.
<i>White ratio</i>	The proportion of applications from White applicants.
<i>Conventional ratio</i>	The proportion of applications for conventional loans. A conventional loan is a mortgage that is not guaranteed or insured by any government agency.
<i>Income average</i>	The natural logarithm of average applicant income in the MSA.
<i>Unemp rate</i>	The unemployment rate in an MSA, multiplied by 100.

Table B.1. (Continued)

Variable	Definitions
<i>Post IPO</i>	An indicator variable equal to one after a state experiences intensive IPO activity, and zero otherwise.
<i>Trade_sector</i>	An indicator variable equal to one if a household works in one of the following 10 industries: agriculture, forestry, and fishing; mining; construction; manufacturing; transportation, communications, electric, gas, and sanitary services; wholesale trade; retail trade; finance, insurance, and real estate; services; and public administration; and zero otherwise.
<i>Income square</i>	The squared natural logarithm of one plus monthly household income.
<i>Industry income</i>	Annual income averaged over the pre- and post-PNTR samples for each industry.
<i>Post_PNTR</i>	An indicator variable equal to one after 2001, and zero otherwise.
<i>Amount_appl</i>	The natural logarithm of the mortgage loan amount in applications.
<i>Hedonic-adjusted house value</i>	Hedonic-adjusted house value is defined as the property value adjusted by hedonic-adjusted price index (Han 2010).
<i>Approval rate</i>	The ratio of the number of granted loans relative to the number of loan applications.
<i>No. appr</i>	The number of mortgage loan approvals in an MSA divided by MSA population.
<i>Property value</i>	The natural logarithm of one plus property value at purchase.
<i>LTV</i>	The ratio of mortgage loans to the house value.
<i>Skill</i>	An indicator variable equal to one if the occupation of the labor head is managerial, professional, or technical, and zero otherwise.
<i>Net inflow</i>	The difference between the total inflow of employees and total outflow of employees.
<i>Total inflow</i>	The number of individuals who move to the current state this year.
<i>Mortgage payment</i>	The natural logarithm of one plus monthly mortgage debt payment.
<i>Payment year</i>	The number of years over which payments are to be made.
<i>Housing supply</i>	Housing supply elasticity: the ratio of relative changes in the inventory of housing to relative changes in the price of housing (Saiz 2010).
<i>Hprice growth</i>	The annual growth rate of house prices in each state.
<i>Manufacturing</i>	An indicator variable equal to one if the industry belongs to manufacturing, and zero otherwise.
<i>Wholesale</i>	An indicator variable equal to one if the industry belongs to wholesale, and zero otherwise.
<i>Personal service</i>	An indicator variable equal to one if the industry belongs to personal service or entertainment service, and zero otherwise.
<i>Agrimining</i>	An indicator variable equal to one if the industry belongs to agriculture or mining, and zero otherwise.
<i>Other industries</i>	An indicator variable equal to one if the industry does not belong to manufacturing, wholesale, personal service, agriculture, or mining, and zero otherwise.

Appendix C. IPO Shock and Local Agglomeration

Table C.1. IPO Shock and Local Agglomeration

Dependent variable	<i>Log(1+mdebt)</i> (1)	<i>Dum_mdebt</i> (2)
<i>Local aggl</i>	0.0415** (2.65)	0.0041*** (3.01)
<i>Post IPO</i>	0.2194** (2.03)	0.0195** (2.06)
Controls	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Observations	34,265	34,265
Adjusted <i>R</i> ²	0.1954	0.1880

Notes. This table presents the result of including both IPO activities and local agglomeration in the regression. *Log(1+mdebt)* in column (1) is the natural logarithm value of a mortgage loan at origination, whereas *Dum_mdebt* in column (2) is a dummy variable indicating if a household has a mortgage loan. Robust *t*-statistics clustered at the state level are reported in parentheses beneath each estimate.

*, **, and *** refer to significance at the 10%, 5%, and 1% levels, respectively.

Endnotes

¹ In 2022, the mortgage debt in the United States amounted to \$19.6 trillion, up from \$18.3 trillion in 2021. See <https://www.federalreserve.gov/apps/FOF/Guide/L217.pdf>.

² In his seminal work, *Principles of Economics*, Marshall (1890) indicated three reasons for the emergence of agglomerated economies: knowledge spillovers, linkages between input suppliers and producers, and labor market interactions. Spillover effects are related to skill acquisition and technology learning, linkages explicitly mention the benefits of sharing intermediate suppliers, and labor market interactions refer to the matching process between workers and job positions.

³ See <https://gustancho.com/mortgage-after-unemployment/>.

⁴ See <https://www.rocketmortgage.com/learn/getting-a-mortgage-with-a-new-job>.

⁵ Just as Glaeser et al. (1992, p. 1127) said, “Intellectual breakthroughs must cross hallways and streets more easily than oceans and continents.” Numerous studies have stressed how agglomerated economies improve the probability of skill learning (Jacobs 1969, Lucas 1988). The advantages of agglomerated economies for learning not only encompass cutting-edge technologies but also the acquisition of skills and “everyday” incremental knowledge creation, diffusion, and accumulation.

⁶ As an example, Detroit used to be one of the most famous clustered industries in the world. However, it filed bankruptcy in 2013, and more than half of the people earning a living in auto-related industries were laid off. The key reason is that Detroit's auto industry had not stepped up its transformation and gradually fell behind other countries' technology.

⁷ To mitigate the concern that the number of mortgage applications and approvals are capturing size effect, we deflate them by local population.

⁸ For example, Mian and Sufi (2011) find that households respond to increased house prices by borrowing more debt, most of which is mortgage debt. They subsequently call this result the home equity-based borrowing channel.

⁹ Within the same local area, local agglomeration measured in our paper varies by the industry in which households work.

¹⁰ The main advantages of SIPP relative to other commonly used data sets, such as the Survey of Consumer Finances and the Panel Study of Income Dynamics, are its large sample size and detailed information about mortgage debts and covariates.

¹¹ We do not focus on outstanding mortgage debt in the panel data structure because it highly follows the payment arrangement, and it reflects more payment behaviors (Cookson et al. 2022).

¹² One may argue that if we only collect the observations for the first year when the household was surveyed, our final sample will be clustered in early sample years. The survey structure of SIPP can mitigate this concern because SIPP provides a short panel (Chetty et al. 2017). Indeed, most households will be surveyed only once during our sample period.

¹³ For the dependent variable $\text{Log}(1+mdebt)$, we estimate a tobit model (left censored at zero), finding that the relation between local agglomeration and mortgage loans is significant at the 1% level. Also, for Dum_mdebt , we continue to find a positive and significant coefficient on local agglomeration if we use probit or logit regression model. We did not use a nonlinear regression model as our main test because it would lead to biased results if we add many fixed effects (Neyman and Scott 1948).

¹⁴ $0.0430 \times 2.252 / 4.780 \times 100\% = 2\%$.

¹⁵ This result means that the negative link between household size and mortgage debt is mainly driven by adult members.

¹⁶ The HMDA data set lacks industry information; therefore, local agglomeration for HMDA data is created at the MSA-year level. Furthermore, to facilitate the interpretation of the coefficient, we multiply it by a factor of 100.

¹⁷ We do not use state-level local agglomeration as our main result as local agglomeration at the MSA level provides more variations than that at the state level. In addition, we do not adopt the HMDA database because HMDA does not provide the detailed demographic information that is necessary for cross-sectional tests.

¹⁸ In Table C.1 of Appendix C, we show that when we add both IPO shock and local agglomeration into the regression, the magnitude and the significance of coefficients on local agglomeration decrease.

¹⁹ The idea behind a stacked DiD is to create event-specific data sets (or cohorts), which include only "clean" controls (e.g., never-treated units and not-yet-treated units) within the treatment window.

²⁰ The results hold when we use different windows, including $[-7, +7]$ and $[-10, +10]$.

²¹ If we randomly assign the IPO shock to state-years, repeat the randomization 500 times, and reestimate our coefficient of interests from 500 bootstrap simulations, we find that the impact of local agglomeration on mortgage loans is statistically indistinguishable from zero.

²² As for housing supply, we add it as a control in alternative explanations (in panel B of Table 8), finding that the positive link between local agglomeration and mortgage loans holds.

²³ The data of house price index in each MSA and year is from the Federal Housing Finance Agency. See <https://www.fhfa.gov/DataTools/Downloads/Pages/House-Price-Index-Datasets.aspx>.

²⁴ Moreover, our results are not sensitive to this cutoff. We find that our results are robust if we exclude observations from markets where local agglomeration ranks in the top 20% or in the top 30%.

²⁵ In untabulated results, we also examine the impact of local agglomeration on mortgage delinquency and find that local agglomeration is negatively associated with delinquency, and the link is significant at the 1% level. This finding therefore further supports the risk-based effect of local agglomeration; that is, households in agglomeration economies have lower default risk.

²⁶ To mitigate the migration concern, we also examine the link between local agglomeration and household mortgage loans in the sample of nonmigrants. We find our results hold when using non-moving households for analysis. Our results also hold if we add migration as an additional control in the regression.

²⁷ To mitigate the concern of house prices, we also limit our sample period to 1984–1999 because house prices in the United States were stable before 1999 (Barrot et al. 2022). We continue to find a significant and positive impact of local agglomeration when using this subsample analysis.

²⁸ The efficient linkage between suppliers and producers, as well as the economies of scale, allows firms to operate productively. For example, studies have shown that firms in urban clusters are more likely to innovate (Glaeser et al. 1992), vertically disintegrate (Holmes 1999), strategically merge (Almazan et al. 2010), and invest efficiently (Dougal et al. 2015).

²⁹ Alternatively, we use the location of firms to measure firm size. Specifically, large firms are defined as firms that operate in multiple locations. Consistent with our first measure, we observe a significantly positive impact of agglomerated economics on households' mortgage loans, regardless of whether they work for multilocalized firms or one-located firms.

³⁰ The key industries listed in Appendix A such as motor vehicles, computers and related equipment, and furniture and fixtures belong to manufacturing.

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